

SEMESTER 3

INFORMATION TECHNOLOGY

SEMESTER S3

MATHEMATICS FOR COMPUTER AND INFORMATION SCIENCE-3

(Group A)

Course Code	GAMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probability and analysis of random processes used in various applications in engineering and science.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, the Binomial probability distribution, the Poisson probability distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Limit theorems : Markov's Inequality, Chebyshev's Inequality, Strong Law of Large Numbers (Without proof), Central Limit Theorem (without proof), Stochastic Processes: Discrete-time process, Continuous-time process, Counting Processes, The Poisson Process, Interarrival times (Theorems without proof) [Text 2: Relevant topics from sections 2.7, 2.9, 5.3]	9
4	Markov Chains, Random Walk Model, Chapman–Kolmogorov Equations, Classification of States, Irreducible Markov chain, Recurrent state, Transient state, Long-Run Proportions. (Theorems without proof) [Text 2: Relevant topics from sections 4.1, 4.2, 4.3, 4.4]	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Familiarize and apply limit theorems and to understand the fundamental characteristics of stochastic processes.	K3
CO4	Solve problems involving Markov Chains, to understand their theoretical foundations and to apply them to model and predict the behaviour of various stochastic processes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introduction to Probability Models	Sheldon M. Ross	Academic Press	13 th edition, 2024

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Random Processes for Electrical and Computer Engineers	John A. Gubner	Cambridge University Press	2012
2	Probability Models for Computer Science	Sheldon M. Ross	Academic Press	1 st edition, 2001
3	Probability, Random Variables and Stochastic Processes	Papoulis, A. & Pillai, S.U.,	Tata McGrawHill.	4 th edition, 2002
4	Probability, Statistics and Random Processes	Kousalya Pappu	Pearson	2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_mg31/preview
2	https://onlinecourses.nptel.ac.in/noc22_mg31/preview
3	https://archive.nptel.ac.in/courses/108/103/108103112/
4	https://archive.nptel.ac.in/courses/108/103/108103112/

SEMESTER S3

COMPUTER ORGANISATION AND ARCHITECTURE

Course Code	PCITT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GXEST203: Foundations of Computing - From Hardware Essentials to Web Design	Course Type	Theory

Course Objectives:

1. Students will be able to gain a comprehensive understanding of the fundamental principles and operational concepts of computer architecture.
2. Students will be able to learn about the key elements and mechanisms of pipelining and its implementation in modern processors.
3. Students will be able to acquire knowledge of cache memory and DRAM systems, focusing on their organisation, performance, and optimisation techniques.
4. Students can explore and differentiate various parallel processing architectures and multicore processors.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals: Basic operational concepts, memory locations and addresses, instructions and instruction sequencing, instruction set principles modes, classifying instruction set architectures, memory addressing, type and size of operands, operations in the instruction set, instructions for control flow, encoding an instruction set, RISC architecture.	11
2	Pipelining: Introduction to single cycle and multi cycle processors. Pipelining- Basic and intermediate concepts, Introduction to pipelining, pipeline hazards, structural hazards, data hazards, control hazards, pipelining implementation, RISC V integer pipeline to handle multicycle operations.	12
3	Memory Organization: Introduction to cache memory, cache mapping, block replacement techniques, cache performance, six basic cache optimizations. DRAM memory system organization, basic nomenclature,	12

	memory modules, basic DRAM memory access protocol, basic DRAM commands, DRAM controller architecture, row buffer management policy, open page row buffer management policy, close page row buffer management policy.	
4	Introduction to Parallel Processors: SISD, MIMD, SIMD, SPMD, and vector hardware multithreading, multicore and other shared memory multiprocessors, Case Study: Intel Core i7.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the fundamental concepts of computer operations, memory organization, and instruction set architectures.	K3
CO2	Describe and apply the concepts of pipelining, including pipeline hazards and their implementation in RISC-V architecture.	K3
CO3	Apply the principles and performance optimization techniques of cache memory and the organization and protocols of DRAM systems.	K2
CO4	Understand different parallel processor architectures and multicore processors, with a case study on Intel Core i7.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	-	-	-	-	-	-	-
CO2	3	3	2	-	-	-	-	-	-	-	-	-
CO3	3	2	3	-	3	-	-	-	-	-	-	-
CO4	3	2	2	-	3	-	-	-	-	2	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Organization and Design	Patterson D.A. and J. L. Hennessey	Morgan Kauffmann Publishers	5 th Edition 2014
2	Computer Organization	Hamacher C., Z. Vranesic and S. Zaky	McGraw Hill	5 th edition 2012
3	Memory Systems: Cache, DRAM, Disk	Bruce Jacob, David T. Wang, and Spencer Ng	Morgan Kauffmann Publishers	1 st Edition 2007
4	Computer Organization- A Quantitative Approach	Patterson D.A. and J. L. Hennessey	Morgan Kauffmann Publishers	6 th Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Organization and Architecture: Designing for Performance	William Stallings	Pearson	9th edition 2013
2	Parallel Computers- Architecture and Programming	V Rajaraman, C Siva Ram Murthy	PHI learning	2nd Edition 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_cs62/preview
2	https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-cs47/

SEMESTER S3

DATA STRUCTURES

Course Code	PCITT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105: Algorithmic Thinking in Python GXEST204: Programming in C	Course Type	Theory

Course Objectives:

1. Students will be able to understand and classify various data structures and their complexities, analyze algorithms for space and time efficiency, and perform basic operations on arrays, stacks, and queues.
2. Students will be able to implement basic searching, sorting, and hash table operations for efficient data management in computational applications.
3. Students will be able to master operations on linked lists and implement stacks and queues using linked lists effectively.
4. Students will be able to achieve a solid grasp of tree and graph structures, their traversal methods, and applications, including the implementation of binary search trees, heaps, and graph algorithms like Dijkstra's shortest path.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Data Structures: Definitions, Classifications of Data Structures- ADT and CDT, Linear and Non-Linear, Static and Dynamic. Algorithm/Program Development: Analysis of Algorithms- Space Complexity, Understanding the notion of Time Complexity using graphical and tabular representation for Linear, Quadratic, Cubical, Logarithmic and Exponential functions. Asymptotic Notations- Best Case, Worst Case, Average Case. Arrays: Definition, Properties of Array. Representation of Single/Two-dimensional Arrays, Sparse Matrix, Polynomial representation, and manipulation using Arrays (Addition only). Stack: Operations on Stack using Array Queues: Definition, Operations on Queue using Array, Circular	11

	Queue implementation using Array, Other types of Queues: Dequeue, Priority Queue (Concept only) Applications of Stack: Infix to Postfix conversion and Postfix evaluation.	
2	Searching: Linear Search and Binary search. Sorting: Bubble Sort, Selection Sort, Insertion Sort. Quicksort and Merge Sort (Recursive Algorithms only) Complexity analysis of sorting algorithms (Detailed analysis is not required) and Heap sort. Hash Tables: Different Hash Functions: Division method, Multiplication Method, Mid Square Method, Folding Method, Collision Resolution Techniques: Closed Hashing (Linear Probing) -Drawbacks, Open Hashing (Separate Chaining). Remedies-Random Probing, Quadratic Probing, Double Hashing. Load Factor and Re-hashing	11
3	Linked List: Definition, Array Vs Linked List, Dynamic memory management. Representation of a Linked List, Classification (Singly Linked List, Doubly Linked List, Circular singly linked list), Operations on Singly Linked List- Traversal, Insertion, Deletion and Searching. Applications of Linked List: Polynomial representation and manipulation (Addition only). Operations on Doubly Linked List- Traversal, Insertion, Deletion and Searching, Circular Linked List (Concept only) Linked List representation of Stack and Queue, Stack implementation and Operations on Stack using Linked List, Queue implementation and Operations on Queue using Linked list.	11
4	Trees: Definition, Height, Depth, Degree, Level. Binary Trees: Properties of Binary Trees, Complete, Perfect and Full Binary Tree, Linear and Linked List representations of Binary Trees, Binary Tree Traversals - Preorder, In order, and Post order (Recursive Algorithms only), Binary Tree Applications: Expression Tree creation. Binary Search Tree: Creation, Insertion, Deletion, and Search operations. Heap: Creation Graph: Definition, set representations, Directed and Undirected Graph, Adjacency Matrix and Adjacency List representation, Graph Traversal: Breadth First Search (BFS), Depth First Search (DFS), Graph Applications: Shortest Path Problem-Dijkstra's Algorithm.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of data structure and analyze algorithm complexity using graphical and tabular methods in implementing stack and queue operations.	K3
CO2	Implement and evaluate searching, sorting, and hash table operations with various collision resolution techniques.	K3
CO3	Implement and manipulate linked lists and utilize them for stack and queue operations, including polynomial manipulation	K2
CO4	Apply tree and graph algorithms to traverse, search, and solve problems, including shortest path computations in graph structures.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	-	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	-	-	-	-	2
CO3	3	3	2	3	2	-	-	-	-	-	-	2
CO4	3	3	3	2	2	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Classic Data Structures	Debasis Samantha	Prentice Hall	2 nd Edition 2008
2	Data Structures Through C	Yashvanth Kanenthkar	BPB Publications	3 rd Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Structures and Algorithms	A. Aho, J. Hopcroft, J. Ullman	Pearson Education	2 nd Edition 2008
2	Introduction to Algorithms	T.H.Cormen, C.E. Leiserson, R.L. Rivest, C. Stein	MIT Press	4th Edition 2022
3	Introduction to Data Structures with Applications	Tremblay J. P. and P. G. Sorenson	Tata McGraw Hill	2 nd Edition 2001
4	Fundamentals of Data Structures in C	E. Horwitz , S. Sahani, D. Mehta	University Press	2 nd Edition 2008

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_cs96/preview

SEMESTER S3

DATABASE MANAGEMENT SYSTEM

Course Code	PBITT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105: Algorithmic Thinking with Python	Course Type	Theory

Course Objectives:

1. Students will be able to understand the elementary concepts of a database management system.
2. Students will be able to design and implement a database project based on these concepts.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: • Overview of a database project and its scope and objectives. • Importance of database management system in real-world applications. Database Design: • Conceptual Design: Create an Entity-Relationship Diagram (ERD) to model the entities, relationships, and attributes involved in your project. • Logical Design: Translate the ERD into a relational schema. Database Creation: • Implement a sample schema in a DBMS of your choice (e.g., MySQL, PostgreSQL, SQLite). Tables and Relationships: • Define tables with appropriate attributes (columns). • Establish relationships (foreign keys) between tables.	9
2	Normalization: • Ensure your schema is normalized to minimize redundancy and dependency issues.	9

	SQL Queries: - Develop a set of SQL queries to demonstrate CRUD operations (Create, Read, Update, Delete), complex joins, aggregations, and any specialized queries required by your project. Timeline: - Create a timeline or project plan with milestones to ensure timely completion.	
3	Constraints: - Entity Integrity - Referential Integrity - Domain Constraints Transaction Management: - Definition of transactions and transaction properties (ACID properties). - Implementation of transaction processing logic. Concurrency Control: - Techniques to manage concurrent access (e.g., locking, time stamping). Backup and Recovery: - Strategies for database backup and recovery in case of failures.	9
4	Testing and Validation: - Execute queries to validate the correctness of the schema and constraints. Integration with Applications: - Develop simple applications (eg; using Python) to interact with the database. Documentation: - Prepare comprehensive documentation covering project objectives, database schema, constraints, transactions, and implementation details.	9

Suggestions on Project Topics:

No.	Project Topics
1	E-commerce Database Platform
2	Registered Vehicle Database Management
3	Library Database Management System
4	Student Database Management
5	Pharmacy Database Management System
6	Restaurant Database Management
7	Payroll Database Management System
8	Insurance Products Database Management
9	Hospital management system.
10	Bank transactions management system.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	- Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a relational database schema for a given project, utilizing Entity-Relationship Diagrams (ERDs) and translating them into a relational schema.	K3
CO2	Normalize a database schema and develop complex SQL queries to manipulate and retrieve data efficiently.	K3
CO3	Implement and manage database constraints, transactions, concurrency control mechanisms, and backup/recovery strategies to ensure data integrity and system reliability.	K3
CO4	Develop database application and prepare detailed documentation.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	2	-	-	-	3	-	3	-
CO2	3	-	3	-	-	-	-	-	3	-	3	-
CO3	3	-	3	-	2	-	-	-	3	2	3	-
CO4	-	-	3	2	2	-	-	-	3	3	3	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Database Systems: Models, Languages, Design and Application Programming	Elmasri R. and S. Navathe	Pearson Education	7 th Edition 2016
2	Database System Concepts	Silberschatz A., H. F. Korth and S. Sudarshan	McGraw Hill	7 th Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Database Systems	C. J. Date, S. Swamynathan and A. Kannan	Pearson Education	8 th Edition 2006
2	Database Management Systems	G.K.Gupta	Tata McGraw Hill	2011

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.swayam2.ac.in/ini24_cs01

Project Based Learning - Course Elements			
L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3
DIGITAL ELECTRONICS AND LOGIC DESIGN

(Common to Group A)

Course Code	GAEST305	CIE Marks	40
Teaching Hours/Week (L:T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Familiarize the basic concepts of Boolean algebra and digital systems. Further, the design of simple combinational and sequential logic circuits which is essential in understanding organization & design of computer systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to digital Systems :- Digital abstraction Number Systems – Binary, Hexadecimal, grouping bits, Base conversion; Binary Arithmetic – Addition and subtraction, Unsigned and Signed numbers; Fixed-Point Number Systems; Floating-Point Number Systems Basic gates- Operation of a Logic circuit; Buffer; Gates - Inverter, AND gate, OR gate, NOR gate, NAND gate, XOR gate, XNOR gate; Digital circuit operation - logic levels, output dc specifications, input dc specifications, noise margins, power supplies; Driving loads - driving other gates, resistive loads and LEDs. Verilog (Part 1) :- HDL Abstraction; Modern digital design flow - Verilog constructs: data types, the module, Verilog operators.	11

2	Combinational Logic Design: – Boolean Algebra - Operations, Axioms, Theorems; Combinational logic analysis - Canonical SOP and POS, Minterm and Maxterm equivalence; Logic minimization - Algebraic minimization, K-map minimization, Dont cares, Code convertors. Modeling concurrent functionality in Verilog:- Continuous assignment - Continuous Assignment with logical operators, Continuous assignment with conditional operators, Continuous assignment with delay.	11
3	MSI Logic and Digital Building Blocks MSI logic - Decoders (One-Hot decoder, 7 segment display decoder), Encoders, Multiplexers, Demultiplexers; Digital Building Blocks - Arithmetic Circuits - Half adder, Full adder, half subtractor, full subtractor; Comparators. Structural design and hierarchy - lower level module instantiation, gate level primitives, user defined primitives, adding delay to primitives.	8
4	Sequential Logic Design :- Latches and Flip-Flops- SR latch, SR latch with enable, JK flipflop, D flipflop, Register Enabled Flip-Flop, Resettable Flip-Flop. Sequential logic timing considerations; Common circuits based on sequential storage devices - toggle flop clock divider, asynchronous ripple counter, shift register. Finite State Machines :- Finite State Machines - logic synthesis for an FSM, FSM design process and design examples; Synchronous Sequential Circuits - Counters; Verilog (Part 2) : - Procedural assignment; Conditional Programming constructs; Test benches; Modeling a D flipflop in Verilog; Modeling an FSM in Verilog.	14

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks. (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the basic concept of different number systems and perform conversion and arithmetic operations between different bases.	K2
CO2	Interpret a combinational logic circuit to determine its logic expression, truth table, and timing information and to synthesize a minimal logic circuit through algebraic manipulation or with a Karnaugh map.	K2
CO3	Illustrate the fundamental role of hardware description languages in modern digital design and be able to develop the hardware models for different digital circuits.	K3
CO4	Develop MSI logic circuits using both the classical digital design approach and the modern HDL-based approach.	K3
CO5	Develop common circuits based on sequential storage devices including counter, shift registers and a finite state machine using the classical digital design approach and an HDL-based structural approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3	3								3
CO3	3	3	3	3	3							3
CO4	3	3	3	3	3							3
CO5	3	3	3	3	3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Logic Circuits & Logic Design with Verilog	Brock J. LaMeres	Springer International Publishing	2/e, 2017
2	Digital Design and Computer Architecture - RISC-V Edition	Sarah L. Harris, David Harris	Morgan Kaufmann	1/e, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Design with an Introduction to the Verilog HDL, VHDL, and System Verilog	M Morris Mano, Michael D Ciletti	Pearson	6/e, 2018
2	Digital Fundamentals	Thomas Floyd	Pearson	11/e, 2015
3	Fundamentals of Digital Logic with Verilog Design	Stephen Brown, Zvonko Vranesic	McGrawHill	3/e, 2014
4	Switching and Finite Automata Theory	Zvi Kohavi Niraj K. Jha	Cambridge University Press	3/e, 2010

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://nptel.ac.in/courses/117105080
2	https://onlinecourses.nptel.ac.in/noc21_ee39/
3	https://onlinecourses.nptel.ac.in/noc24_cs61/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP_{fc} from GDP_{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP_{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, $NFIA=(-200)$ and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics .	6

	Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6

4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

PROGRAMMING IN PYTHON LAB

Course Code	PCITL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105: Algorithm Thinking With Python	Course Type	Lab

Course Objectives:

1. Students will be able to develop skills in text processing, advanced function handling, and exception handling.
2. Students will be able to provide hands-on experience with image processing techniques and object-oriented programming principles.

SYLLABUS

Experiment No.	Experiments
1	a) Creation of List & List Operations b) Advanced List Comprehension c) Tuple and Tuple operations d) Creation of Dictionary and Operations
2	Text Processing Sample Program a) Given an input file which contains a list of names and phone numbers separated by spaces in the following format: "Phone Number contains a 3- or 2-digit area code and a hyphen followed by an 8-digit number". Find all names having phone numbers with a 3-digit area code using regular expressions. b) Write a Python program to check the validity of a password given by the user. c) Write a Python program to validate mobile number

3	Working with functions Sample Program: a) Write a program to solve a classic ancient Chinese puzzle: Get the input from the user for the number of heads (eg: 35 heads) and legs (eg: 94 legs) among the chickens and rabbits in a farm. Display the number of rabbits and chickens we have. Use the function solve(heads, legs). Hint: Use for loop to iterate all possible solutions. b) You are given an array A of size N. Your friend gave me an amazing task for you. Your friend likes one type of Sequence. So, he called that type of sequence a fair sequence. You should select a fair sequence of maximum length from an array. Here a fair sequence is nothing but you have to select elements in a pattern like positive element, negative element, positive element... negative element, positive element, negative element, to form a sequence. Your task is to print the maximum sum of elements possible by selecting a fair subsequence with maximum length. (Use functions). Ex: If array A = [-1, 18, 13, 18, 2, 16, 7, -1, -213, 11]. Here your minimum length can be 6. The fair subsequence is -1, -18, -7, -2, 7, -1, 11. The Sum is 32 which is the maximum possible. Your friend also kept a timer of 15 min. You will win, will you be able to do it? c) Working with Lambda function
4	Exception Handling and User defined exception(s) Sample Program: a) Write a python program to catch different types of exceptions b) Write a python program to create user defined exceptions.
5	File handling Sample Program: Write a Python program that manages book records using a binary file called book.dat. The program should have the following functions: • add_record() This function should prompt the user to input data for a book, including the Book Number, Book Name, Author, and Price. The data should be stored in a list format:

	[BookNo, Book_Name, Author, Price]. The function should then add the book data to the book.dat binary file using pickling. ● display_records() This function should display all the records stored in the book.dat file on the screen. Each record should be displayed with its corresponding attributes: Book Number, Book Name, Author, and Price. ● books_by_author() This function should ask the user to input an author's name. It should then search through the records in the book.dat file and display all the records for books written by the given author. ● books_by_price(price) This function should receive a maximum price as an argument. It should search through the records in the book.dat file and display all the records for books with a price less than the provided maximum price. ● copy_data() This function should read the contents of the book.dat file. It should copy the records of books whose price is more than \$500 to a new file named costly_book.dat using pickling. The function should return the count of records copied. ● delete_record(book_number) This function should receive book number as an argument. If a record with the given book number exists, it should be deleted from the book.dat file. If the record does not exist, display an error message. ● update_record(book_number) This function should receive a maximum price as an argument. If a record with the given book number exists, prompt the user to update the Book Name, Author, and Price for that record. If the record does not exist, display an error message. In the program, include a menu-based interface that allows users to choose which function to execute.
6	Working with data (Use of user- defined/ Built-in modules) Sample Program: a) Querying an SQL light database- insert, update, select, delete

	b) Comparing data in a .CSV file and write to a new .CSV file.
7	Working with data (Use of user- defined/ Built-in modules): Getting data from Web, JSON Sample Program: a) Scrap HTML content from a page and pass the code with beautiful soup. b) Serializing/ Deserializing JSON – Access and work with data stored as JSON
8	Image Processing Sample Program: a) Reading, Saving and displaying an image using OpenCV-PyPI, matplotlib
9	Advanced Image processing Sample Program: a) Image statistics cropping, b) Converting images from RGB to Gray and Resizing the image c) Skin color detection
10	Basic Object-Oriented Sample Program: a) Write a Python class named Person with attributes name, age, weight (kgs), height (ft) and takes them through the constructor and exposes a method get_bmi_result() which returns one of "underweight", "healthy", "obese". b) Write a python program to demonstrate operator overloading.
11	Advanced Object-Oriented Sample Program: a) Write a python program to demonstrate various kinds of inheritance b) Write a python program to create abstract classes and abstract methods
12	Implement a micro project using any of the Python concepts described in experiments 1-11. For Projects, Students may be grouped with maximum of 4 persons. Evaluation shall be done on demonstration and presentation of project for maximum 10 minutes.

	Example: Creating a snake and ladder game, PageRank Algorithm implementation etc.
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Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply advanced list comprehensions, tuples, and dictionaries for efficient data manipulation and develop text processing scripts, including functions with advanced argument handling and lambda functions.	K3
CO2	Implement robust programs using exception handling and user-defined exceptions.	K3
CO3	Implement file operations, work with relational databases, CSV, and web-based data sources effectively.	K3
CO4	Perform basic and advanced image processing using libraries such as OpenCV and matplotlib, and demonstrate basic and advanced object-oriented programming concepts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	3	-	-	--	-	-	3	3	-	2
CO2	3	-	3	-	-	-	-	-	3	3	-	2
CO3	3	-	3	-	-	-	-	-	3	3	-	2
CO4	3	-	3	2	3	-	-	-	3	3	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	How to think like a Computer Scientist- Learning with Python	Allen Downey, Jeffrey Elkner, Chris Meyers	Green Tea Press	1 st edition, 2002
2	Learning Python: Powerful Object-Oriented Programming	Mark Lutz	O'Reilly Media Inc.	5 th edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Problem Solving and PYTHON Programming	S.A.Kulkarni	Yes Dee Publishing Pvt Ltd	2 nd edition, 2008
2	Programming in Python 3: A Complete Introduction to the Python Language	Mark Summerfield	Pearson Education	2 nd edition, 2018
3	Fundamentals of Python	Kenneth A. Lambert, B. L. Juneja	Cengage Learning India Pvt. Ltd.	1 st edition, 2015
4	Let Us Python	Yashavant Kanetkar ,Aditya Kanetkar	BPB Publications	1 st edition, 2019

Video Links (NPTEL, SWAYAM...)	
Link No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_cs40/preview
2	www.https://realpython.com/
3	https://onlinecourses.nptel.ac.in/noc24_cs113/preview
4	https://docs.python.org/3/reference/

Continuous Assessment (25 Marks):

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks):

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related question.
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

DATA STRUCTURES LAB

Course Code	PCITL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GXEST204: Programming in C	Course Type	Lab

Course Objectives:

1. Understand the importance of data structures, abstract data type, and their basic usability in different applications.
2. Implement basic searching, sorting, and hashing algorithms to efficiently solve data organization and retrieval problems.
3. Implement linear and non-linear data structures using arrays and linked lists.
4. Apply various data structures such as stacks, queues, trees, graphs, etc. to solve various computing problems.
5. Identify suitable data structures and algorithms to solve a real-world problem.

Experiment No.	Experiments
1	Implementation of Stack using array.
2	Utilize stack to perform the following tasks: i) Convert an infix expression to a postfix expression. ii) Evaluate a given postfix expression
3	Representation of polynomials using arrays and perform polynomial addition.
4	Implementation of Queue and Circular Queue using arrays.
5	Implementation of various sorting algorithms – Bubble sort, Insertion sort, Selection sort, Quick sort, and Merge sort.
6	Implementation of basic searching algorithms – linear search and binary search.

7	Implementation of hash tables using various mapping functions and resolving collisions (if any) using various collision resolution schemes.
8	Implementation of Singly linked list operations- traversing, searching, insertion, deletion.
9	Implementation of Doubly linked list operations- traversing, searching, insertion, and deletion.
10	Representation of polynomials using linked lists and perform polynomial addition.
11	Implementation of Stack and Queue using linked list.
12	Implementation of preorder, in-order, and post-order traversals on binary trees using recursive algorithm.
13	Implementation of binary search trees – creation, insertion, deletion, searches.
14	Implementation of heap sort.
15	Representation of graph using adjacency list and adjacency matrix and compute various parameters (in degree, out-degree, etc.).
16	Implementation of BFS and DFS for each graph representation.
17	Implementation of Dijkstra's algorithm for finding the shortest path.
18	Any application program using trees.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Perform operations on fundamental data structures such as arrays, stacks, queues, linked lists, and hash tables.	K3
CO2	Implement and analyze various sorting (e.g., bubble, insertion, selection, quick, merge) and searching algorithms (linear and binary search), understanding their efficiency and use cases.	K3
CO3	Solve complex problems involving tree and graph data structures.	K3
CO4	Analyze and optimize the performance of data structure operations, including polynomial representation and addition, using both arrays and linked lists and different implementations of stacks and queues.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	2	-	-	-	-	-	-	1
CO2	3	3	2	2	2	-	-	-	-	-	-	2
CO3	3	3	3	2	2	-	-	-	-	-	-	2
CO4	3	3	2	3	3	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Classic Data Structures	Debasis Samantha	Prentice Hall	2 nd Edition 2008
2	Data Structures Through C	Yashvanth Kanenthkar	BPB Publications	3 rd Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Structures and Algorithms	A. Aho, J. Hopcroft, J. Ulman	Pearson Education	2 nd Edition 2008
2	Introduction to Algorithms	T.H.Cormen, C.E. Leiserson, R.L. Rivest, C. Stein	MIT Press	4th Edition 2022
3	Introduction to Data Structures with Applications	Tremblay J. P. and P. G. Sorenson	Tata McGraw Hill	2 nd Edition 2001
4	Fundamentals of Data Structures in C	E. Horwitz , S. Sahani, D. Mehta	University Press	2 nd Edition 2008

Video Links (NPTEL, SWAYAM...)	
Link No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_cs96/preview

Continuous Assessment (25 Marks):

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.

- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks):

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related question.
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

INFORMATION TECHNOLOGY